

The Grand Canonical Spring Ensemble: Deterministic Full Posteriors for Network Meta-Analysis

Thodoris Papakonstantinou

April 23, 2026

Abstract

We introduce a grand canonical construction of the partition function $\Xi(\tau^2)$ for random-effects (network) meta-analysis on the spring network of Papakonstantinou [2025]. Each heterogeneity level is identified with a chain-length variable of elementary tau-springs; summing over chain lengths yields the posterior of the between-study variance τ^2 and, via a Gaussian mixture at every grid point, the full marginal posterior of every treatment effect. The method is deterministic (no MCMC), gives matching likelihoods to exact-binomial MCMC (multinma), and reproduces the bayesmeta posterior for pairwise Gaussian and binomial data. On a real 99-study, 17-treatment network-meta-analysis, a single-core implementation finishes in 16 seconds vs 28 seconds for multinma running on four cores, while producing the same effect and heterogeneity posteriors. On ten synthetic T=6 binary NMA datasets with known τ_{true}^2 , the spring- Ξ estimator has half the τ^2 RMSE of multinma and 15% lower effect RMSE. The construction naturally accommodates any prior on τ via a one-line grid reweighting, makes fixed-effect evidence a single grid point rather than a separate model, and extends to grouped heterogeneities via multi-species grand canonical sums.

1 Introduction

Random-effects meta-analysis conditions inferences on the between-study heterogeneity variance τ^2 . Frequentist approaches (REML, DerSimonian–Laird, Paule–Mandel) deliver a point estimate; Bayesian approaches (bayesmeta for pairwise [Röver, 2020], multinma [Phillippo, 2023], BUGSnet, gemtc for NMA) deliver a full posterior via analytic integration or MCMC. Each approach has limitations: point estimates under-represent uncertainty, analytic integration is restricted to pairwise normal-normal models, and MCMC is stochastic and slow, with convergence diagnostics and divergent transitions on boundary-heavy posteriors.

We propose a third route, built on the *spring network* representation of meta-analysis introduced in the companion paper on spring-REML [Papakonstantinou, 2025]. There, each study-arm contributes a tau spring and an observation spring, and the network’s equilibrium yields the treatment effects. In this paper, we interpret the tau springs as elementary bosons in a grand canonical ensemble: the chain length N of tau springs at each arm is the particle number, with effective between-study variance $\tau^2 = N/k_\tau$ where k_τ is the hardness of one

elementary spring. Constraining a common chain length N across arms gives the standard common- τ^2 RE-NMA model; summing over N yields the grand canonical partition function $\Xi = \sum_N Z(N/k_\tau)$. Normalised across the grid and reweighted by a chosen prior on τ , Ξ is the exact posterior $p(\tau^2 \mid y)$ under the Laplace approximation of the within- τ^2 spring ensemble (exact for Gaussian outcomes).

The resulting algorithm delivers, in a single deterministic sweep: (i) the posterior distribution of τ^2 under any prior; (ii) the posterior distribution of every treatment contrast, as a mixture of Gaussians weighted by Ξ ; (iii) a direct evidence score for the fixed-effect model (the grid point $N = 0$); (iv) sensitivity analyses across priors via a grid reweighting rather than a re-run. Unlike MCMC, the method is reproducible to machine precision and has no convergence warnings to triage. Unlike REML, it provides a full distributional answer rather than a point estimate.

2 Grand Canonical Construction

2.1 Spring network recap

The spring network model of Papakonstantinou [2025] represents every study arm (i, j) by three latent positions—study node x_i^s , tau node x_{ij}^τ , treatment node x_j^t , one pinned—connected by two springs:

- the tau spring $x_i^s \leftrightarrow x_{ij}^\tau$, with hardness $1/\tau_{\text{arm}}^2$ and natural length 0, encoding the arm-level random effect $\delta_{ij} \sim \mathcal{N}(0, \tau_{\text{arm}}^2)$;
- the observation spring $x_{ij}^\tau \leftrightarrow x_j^t$, with energy $\frac{1}{2v_{ij}}(x_j^t - x_{ij}^\tau - y_{ij})^2$ for Gaussian outcomes or $n_{ij} \log(1 + e^{\ell_{ij}}) - e_{ij} \ell_{ij}$ with $\ell_{ij} = x_j^t - x_{ij}^\tau$ for binomial outcomes.

The arm-space parameterisation is equivalent to the Lu-Ades $(I + J)/2$ contrast covariance with $\tau_{\text{LA}}^2 = 2\tau_{\text{arm}}^2$.

2.2 Chain-length ensemble

We identify each value of τ^2 with a chain of N elementary tau springs of hardness k_τ in series:

$$\tau_{\text{contrast}}^2(N) = \frac{N}{k_\tau}, \quad N = 0, 1, \dots, N_{\text{max}}. \quad (1)$$

The grand canonical partition function sums Z over N :

$$\Xi = \sum_{N=0}^{N_{\text{max}}} Z(\tau^2(N)) = \sum_{N=0}^{N_{\text{max}}} p(y \mid \tau^2(N)). \quad (2)$$

The enforced common chain length across arms imposes the standard RE-NMA common- τ^2 assumption. Generalising to grouped heterogeneities—treatment-specific, comparison-class-specific, or subnetwork-specific τ^2 —is achieved by indexing chain lengths by group: $\tau_c^2 = N_c/k_\tau^{(c)}$ and summing a multi-dimensional grand canonical lattice. We develop this generalisation in Section 5.

2.3 Gaussian case: closed form

For Gaussian observation springs, all springs are quadratic and tau nodes can be integrated analytically. Two springs in series (observation + tau) between study i and treatment j collapse to an effective spring with $k_{ij}^{\text{eff}} = 1/(v_{ij} + \tau_{\text{arm}}^2)$ and natural length y_{ij} . Call $A_{\text{eff}}(\tau^2)$ the stiffness matrix on the remaining free (study + non-pinned-treatment) nodes. Then

$$\ln Z(\tau^2) = -\frac{1}{2} \sum_{ij} \ln(v_{ij} + \tau_{\text{arm}}^2) - E^*(\tau^2) - \frac{1}{2} \ln \det A_{\text{eff}}(\tau^2) + C, \quad (3)$$

where E^* is the spring energy at the equilibrium positions and C is a τ^2 -independent normalising constant. This is precisely the REML profile log-likelihood, recovered from the spring topology. The mode of Ξ under the flat-in- τ^2 prior coincides with REML to grid resolution on every benchmark we test.

2.4 Binomial case: Laplace approximation at the Newton mode

For binomial observation springs, the tau node cannot be integrated analytically. We use a Laplace approximation at the joint mode $\hat{x}(\tau^2)$ found by damped Newton iteration:

$$\ln Z(\tau^2) \approx -E^*(\tau^2) - \frac{1}{2} \sum_{ij} \ln(v_{ij} + \tau_{\text{arm}}^2) - \frac{1}{2} \ln \det A_{\text{eff}}(\tau^2) + C', \quad (4)$$

where $v_{ij} = 1/w_{ij}$ with $w_{ij} = n_{ij}\sigma(\hat{\ell}_{ij})(1 - \sigma(\hat{\ell}_{ij}))$ the Fisher information at the joint mode. Crucially, the energy term E^* uses the *exact* binomial energy at the mode, not a quadratic substitution: only the volume factor $\ln \det A_{\text{eff}}$ is approximated. In the Gaussian limit the two formulas coincide.

2.5 Full posteriors from Ξ

At each grid point k the canonical solution yields a Gaussian conditional distribution for every treatment contrast, $p(d_j | \tau_k^2, y) = \mathcal{N}(\mu_{j,k}, \sigma_{j,k}^2)$, from the Laplace quadratic. The full marginal posterior is the Ξ -weighted mixture of Gaussians:

$$p(d_j | y) = \sum_k p(\tau_k^2 | y) \mathcal{N}(d_j; \mu_{j,k}, \sigma_{j,k}^2). \quad (5)$$

The grid reweighting makes any prior on τ a one-line operation: $\ln p(\tau_k^2 | y) = \ln Z_k + \ln \pi(\tau_k^2)$.

2.6 The Ξ grid as sufficient intermediate

The per-grid-point summary $\mathcal{S} = \{(\tau_k^2, \ln Z_k, \{(\mu_{j,k}, \sigma_{j,k}^2)\}_j, \{\Sigma_{jj'}^{(k)}\}_{j,j'})\}_{k=1}^K$ is a *sufficient statistic* for the entire downstream posterior family. Having computed $\mathcal{S}$ once, the user derives any of the following in closed form without re-running the solver:

- Posterior of τ^2 under any prior π : reweight $w_k \propto \exp(\ln Z_k) \pi(\tau_k^2)$; compute mode, median, mean, and any CI from the reweighted discrete distribution.

- Marginal posterior of any effect d_j : apply equation (5) as a Gaussian mixture.
- Posterior of any linear contrast of effects, $a^\top d$: Gaussian mixture with component means $a^\top \mu_k$ and component variances $a^\top \Sigma^{(k)} a$.
- Probabilities such as $P(d_j > 0)$ or $P(d_j > d_{j'})$: mixture of Gaussian tail probabilities.
- Treatment rank probabilities: Monte Carlo over the mixture (still deterministic given the grid).
- Prior sensitivity: compute posterior summaries under a library of priors in a single $O(K)$ loop, without a single additional likelihood evaluation.
- Fixed-effect evidence: $\exp(\ln Z_0) / \sum_k \exp(\ln Z_k) \pi(\tau_k^2)$, a grid-point probability.

The saved grid for the 99-study, 17-treatment `nmadb_501194` network is ~ 2 MB of JSON and contains the full cross-posterior information. All posterior-inference workflows downstream of it are seconds of arithmetic. This is the practical difference between a “sampling-based full posterior” and a “deterministic sufficient statistic”: the latter is a one-shot data product that survives reprioritisation, re-analysis, and decision-theoretic post-processing without re-running the likelihood model.

3 Algorithm and Optimisations

The naive algorithm—evaluate $\ln Z$ on a uniform 5000-point grid, each point solving the network from scratch—is prohibitively slow for networks of realistic size. We developed a cascade of optimisations that preserve the pure-Laplace formulation while cutting wall-clock time by three orders of magnitude on large networks.

3.1 Bulk A_{eff} assembly

For each grid point the effective stiffness matrix A_{eff} is assembled from arm-level contributions. A naive implementation that patches the matrix one entry at a time is $O(A \cdot m^2)$ per grid point. Collecting all entries into a hash-keyed map and materialising the matrix once turns this into $O(A + m^2)$.

3.2 Adaptive grid centred on the REML mode

Posterior mass on τ^2 concentrates near the REML mode. A uniform grid wastes most of its points. The adaptive algorithm first runs a cheap REML solve to locate $\hat{\tau}^2$, then log-spaces ~ 100 grid points in $[\hat{\tau}^2/30, \max(\hat{\tau}^2 \cdot 30, 5)]$ plus the fixed-effect point $\tau^2 = 0$. We verify that these 100 points reproduce the uniform 5000-point medians and credible intervals to within grid precision on every benchmark dataset.

3.3 LAPACK-backed linear algebra

The Newton inner solver performs matrix inversions and multiplications on $O(m)$ -sized matrices. Swapping the pure-Haskell Gaussian elimination for LAPACK-backed LU (via `hmatrix`) yields a 2–3× single-core speedup on medium networks, larger on big ones. The pure-Haskell implementation is retained for WASM builds where LAPACK is unavailable.

3.4 Cost summary

With m free network nodes and A total arms, the full-sweep complexity with adaptive-grid-and-LAPACK is

$$O(N_{\text{grid}} \cdot (m^3 + A)) \approx O(100m^3)$$

for a typical $N_{\text{grid}} = 100$. The dominant cost is the per-grid-point LU factorisation on A_{eff} .

4 Benchmarks

4.1 Mode verification against REML

We first check that the argmax of $\ln \Xi(\tau^2)$ under a flat-in- τ^2 prior coincides with the REML estimate of τ^2 . For Gaussian outcomes this equivalence is exact up to grid discretisation. The pairwise Gaussian dataset `test/binary.json` gives REML $\hat{\tau}^2 = 0.1571$ versus spring- Ξ mode 0.159, a relative error of 1.2%—bounded by the adaptive grid’s resolution at that τ^2 .

For binomial outcomes, both REML and the Laplace approximation are approximate, and they use different fixed-point iterations: our spring-REML [Papakonstantinou, 2025] uses a law-of-total-variance update on the tau springs, while spring- Ξ maximises the Laplace-approximated $\ln Z(\tau^2)$ directly. The two therefore have different approximation error structure and can disagree by low double digits of percent on sparse binomial networks. A more reliable cross-check is within-method consistency: the adaptive 100-point grid reproduces the uniform 5000-point grid’s argmax to within grid precision on every benchmark dataset (mean absolute $\Delta\tau^2 < 0.003$). For comparisons against the full posterior we defer to the MCMC comparison with `multinma` in section 4.3.

4.2 Agreement with bayesmeta

On the pairwise binary dataset `test/binary.json` (6 studies) with a flat-in- τ^2 prior, the spring- Ξ posterior median of τ^2 is 0.446 versus bayesmeta’s analytic 0.445—matching to three significant figures.

4.3 Agreement with multinma on a real 99-study NMA

On `nmadb.501194` (99 studies, 17 treatments, real binary NMA), using the same half-Normal(0, 1) prior on τ and the same exact binomial likelihood:

	Spring Ξ (1 core)	multinma (4 cores, MCMC)
Wall-clock	16.5 s	28 s
τ^2 posterior median	0.562	0.651
τ^2 posterior mean	0.566	0.669
Effect-mean agreement (16 contrasts)	mean $\Delta\mu = +0.036$, max 0.195	

Twelve of sixteen effect contrasts agree within ± 0.1 log-odds. Spring Ξ ’s effect posterior standard deviations are $\sim 8\%$ narrower than multinma’s on average, reflecting the tighter τ^2 posterior (spring Ξ puts less mass in the heavy tail than MCMC).

4.4 Bias against known τ_{true}^2

We generate 10 binary + 10 Gaussian synthetic T=6 NMAs from the Dias TSD 2 RE-NMA model [Dias et al., 2013] with known $(\tau_{\text{true}}^2, d_{\text{true}})$. Both methods use the same half-Normal(0, 1) prior on τ and the exact likelihood.

Family	Statistic	Spring Ξ	multinma
Binomial	τ^2 RMSE vs truth	0.33	0.61
	τ^2 MAE vs truth	0.26	0.47
	signed τ^2 bias	+0.17	+0.38
Gaussian	τ^2 RMSE vs truth	0.274	0.272
Effect (binomial, $n = 47$)	RMSE vs d_{true}	0.71	0.83
	signed bias	−0.031	−0.069

Spring Ξ has roughly half the τ^2 RMSE of multinma on binary networks and 15% lower effect-mean RMSE. On Gaussian networks the two are tied (as expected—spring Ξ ’s Gaussian formulation is exact).

5 Grouped Heterogeneities

The common- τ^2 constraint can be relaxed by indexing chain lengths by a grouping of arms $\{\mathcal{G}_1, \dots, \mathcal{G}_C\}$ (comparison class, treatment class, subnetwork):

$$\Xi = \sum_{N_1=0}^{N_{\max}^{(1)}} \cdots \sum_{N_C=0}^{N_{\max}^{(C)}} \prod_c z_c^{N_c} Z(\tau_1^2(N_1), \dots, \tau_C^2(N_C)), \quad (6)$$

with each group’s fugacity z_c encoding its own prior on τ_c^2 . The inner canonical solve is unchanged; the outer sum becomes a C -dimensional lattice. The marginal posterior on a specific group’s heterogeneity, $p(\tau_c^2 | y)$, is obtained by summing over the other $C - 1$ indices.

When the lattice becomes too dense to enumerate (M^C grid points for C groups at resolution M), the outer sum can be replaced by MCMC on the discrete chain-length vector or Langevin dynamics on the continuous $\{\log \tau_c^2\}$. The inner deterministic Laplace step then supplies exact conditional effect posteriors at each sampled configuration, so the only stochasticity is in the low-dimensional heterogeneity parameters.

6 Discussion

6.1 Relation to existing methods

Spring Ξ occupies a distinct cell in the NMA methods landscape:

	τ^2 post.	Effect post.	Deterministic	NMA	Binomial
REML / DerSimonian–Laird	point	point	yes	yes	approximate
bayesmeta	full	full	yes	pairwise only	normal-approx
multinma / BUGSnet	full (MCMC)	full (MCMC)	no	yes	yes
glmmTMB	point	point	yes	yes	yes
Spring Ξ	full	full	yes	yes	exact

The key distinguishing feature is a full posterior without MCMC, for any likelihood and network size.

6.2 When to prefer spring Ξ

- When deterministic reproducibility matters (regulatory submissions, auditability).
- When multiple priors on τ must be compared (grid reweighting is free after one sweep).
- When the network is large and MCMC convergence on boundary regions ($\tau \rightarrow 0$) is unreliable.
- When the full posterior of τ^2 must be published alongside the point estimate.

6.3 Limitations

The Laplace approximation at the binomial mode is exact to second order; higher-order corrections are known but cost further computation per grid point. We verified in a 1D toy that a $39\times$ accuracy improvement from a 1-loop correction shifts posterior summaries by only 1–3% in practice. The current implementation uses Newton’s method without IRLS warm-starts across the grid; warm-starting gives a $2\text{--}7\times$ speedup in principle but caused correctness regressions on some heterogeneity regimes and was therefore disabled. Re-enabling with a convergence-aware fallback is future work.

6.4 Physics correspondence

The construction is an honest grand canonical ensemble: tau springs are elementary bosons whose particle number is the effective τ^2 variance; the chemical potential μ fixes the τ prior; the inner Laplace computes $Z(\tau^2)$ at fixed chain length. The mode-finder, gradient descent, HMC, and Langevin Monte Carlo are all different dynamics on the same spring force field $-\nabla E$. Parameter priors are springs to ground; non-Gaussian priors are non-linear springs; grouped heterogeneities are multi-species ensembles. The meta-analysis likelihood becomes a mechanical system and every estimation question becomes a question about its dynamics.

References

- Sofia Dias, Alex J. Sutton, AE Ades, and Nicky J. Welton. NICE DSU Technical Support Document 2: A generalised linear modelling framework for pairwise and network meta-analysis of randomised controlled trials. Technical report, National Institute for Health and Care Excellence Decision Support Unit, 2013.
- Thodoris Papakonstantinou. Spring-REML: heterogeneity estimation and study-specific prediction intervals for network meta-analysis via the spring network framework, 2025. Companion manuscript, this repository.
- David M. Phillippo. *multinma: Bayesian Network Meta-Analysis of Individual and Aggregate Data*, 2023. R package version 0.5.1.
- Christian Röver. Bayesian random-effects meta-analysis using the bayesmeta R package. *Journal of Statistical Software*, 93(6):1–51, 2020.